

- 4 Fig. 4.1 shows the structure of the R groups of three amino acids, aspartate (Asp), glutamine (Gln) and isoleucine (Ile).

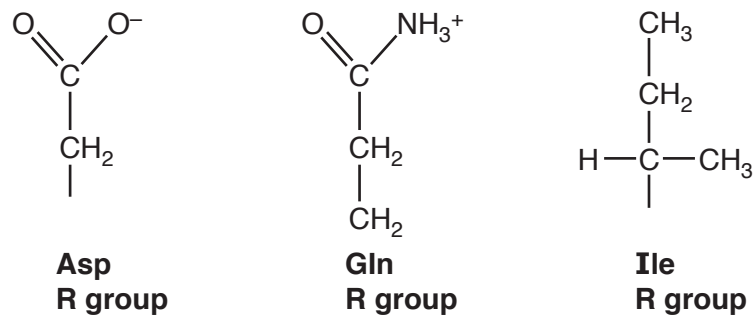


Fig. 4.1

- (a)** One of the mRNA codons for the amino acid Gln is CAA. A mutation could change this codon from CAA to UAA.

UAA is a STOP codon.

State what is meant by a STOP codon.

.....[1]

- (b)** Many proteins that contain the amino acids shown in Fig. 4.1 are globular.

Describe how the three R groups in Fig. 4.1 can contribute to the globular structure of a protein.

[4]

- (c) An example of a globular protein is the enzyme starch phosphorylase. This enzyme catalyses the conversion of starch to glucose-1-phosphate.

Name the type of bond that is broken when starch is converted to glucose-1-phosphate.

.....[1]

The results are shown in Fig. 4.2.



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(e) State how the Michaelis–Menten constant (K_m) is derived from V_{max} for an enzyme.

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.....[1]

(f) The K_m values for starch phosphorylase were determined when free in solution and when immobilised.

The K_m values were:

- free in solution, $K_m = 0.34 \text{ mmol dm}^{-3}$
- immobilised, $K_m = 0.84 \text{ mmol dm}^{-3}$.

Suggest an explanation for the difference between the two values.

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.....[2]

[Total: 13]